

4] Linear partial Differential Equation with constant coefficients.

Linear non-homogeneous higher order PDE is in the following form:-

$$(A_0 D^{n-1} + A_1 D^{n-2} D' + \dots + A_n D'^n) z = f(x, y)$$

$$F(D, D') z = f(x, y) \text{ or } F(D_x, D_y) z = f(x, y).$$

* Solution of LPDE:

put $D = D_x = m$ & $D' = D_y = 1$ in homogeneous part

* Let, $z = \phi(y + mx)$ be a trial soln.

* Distinct Roots: If $z = m_1, m_2, m_3, \dots, m_n$ then.

$$z = \phi_1(y + m_1 x) + \phi_2(y + m_2 x) + \dots + \phi_n(y + m_n x)$$

* Equal / Repeated Roots:

$$z = \phi_1(y + m_1 x) + x \phi_2(y + m_1 x) + x^2 \phi_3(y + m_1 x) + \dots$$

* If $F(Dx, Dy) = f(x, y)$ then

$$Z = \frac{1}{F(Dx, Dy)} \phi(ax + by).$$

$$(Z) = \frac{1}{F(a, b)} \iiint \dots \iint v dv dv \dots dv$$

[where $v = ax + by$]

☐ Find the following PDE's with constant coefficients.

1]. $(D_x^2 - 2D_x D_y + D_y^2)Z = e^{3x-2y}$

2]. $(D_x^3 - 4D_x^2 D_y + 4D_x D_y^2)Z = \cos(x+3y)$

3]. $(D_x^2 - 5D_x D_y + 4D_y^2)Z = \sin(4x-2y)$

4]. $x + y - z = (2x+y)^{\frac{1}{2}}$

5]. $(D_x^3 - 6D_x^2 D_y + 11D_x D_y^2 - 6D_y^3)Z = e^{5x+6y}$

6]. $(D_x^3 - 4D_x^2 D_y + 4D_x D_y^2)Z = 6\sin(3x+2y)$

7]. $(2D_x^2 - 5D_x D_y + 2D_y^2)Z = 24(y-x)$

Soln:-3 Given, (i) $(D_x^2 - 5D_x D_y + 4D_y^2)z = \sin(4x+2y)$ — ①

Let, (ii) $(D_x^2 - 5D_x D_y + 4D_y^2)z = 0$ — ②

Also Let, $z = \phi(y+mx)$ be a trial soln of (ii),

put, $D_x = m$ & $D_y = 1$ in (ii)

$$\therefore (m^2 - 5m + 4)\phi(y+mx) = 0$$

Since, $\phi(y+mx) \neq 0$, Auxilliary eqn will be,

$$m^2 - 5m + 4 = 0$$

$$\Rightarrow m^2 - 4m - m + 4 = 0$$

$$\Rightarrow (m-4)(m-1) = 0$$

$\therefore m = 1, 4$ (Distinct Roots)

$$\therefore z_1 = \phi_1(y+x) + \phi_2(y+4x)$$

Also from (i), $z_2 = \frac{1}{D^2x - 5Dx + 4D^2y} \sin(4x+2y)$

$$(ii) \quad \frac{1}{D^2x - 5Dx + 4D^2y} \sin(4x+2y)$$

$$= \frac{1}{4^2 - 5 \times 4 \times 2 + 4 \times 2^2} \int \sin v \, dv$$

$$= \frac{1}{16 - 40 + 16} \int (-\cos v) \, dv$$

$$= \frac{-1}{-8} \sin v$$

$$= \frac{1}{8} \sin(4x+2y)$$

$\therefore$ General soln is: $z = z_1 + z_2$

$$\therefore z = \phi_1(y+x) + \phi_2(y+4x) + \frac{1}{8} \sin(4x+2y).$$

Ans